
accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000

IBM's Mobile E-business Architecture

Kiran Mutukuru
Senior IT Specialist, IBM

PalmSource
Conference 2000

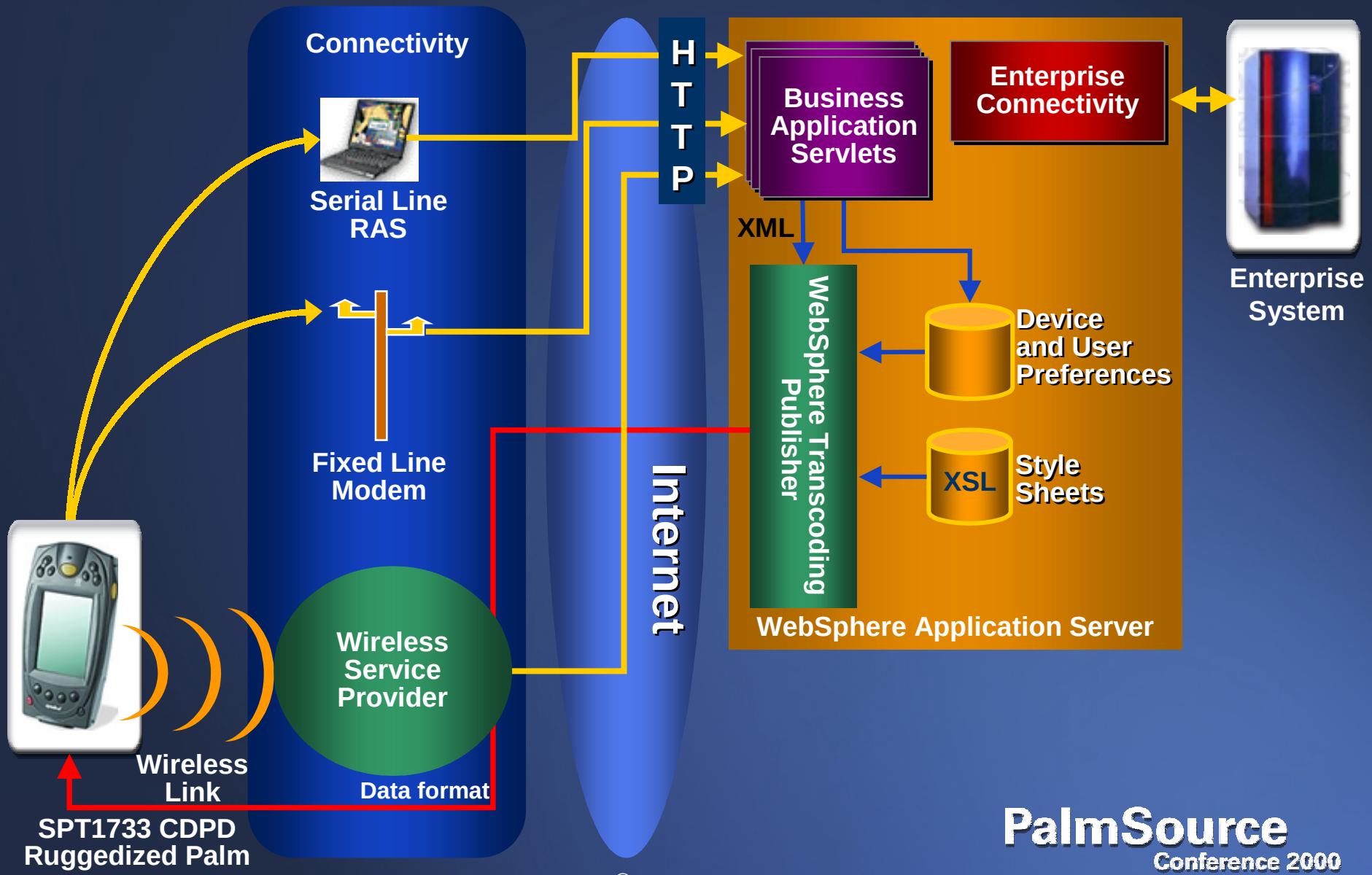
Agenda

- IBM's mobile e-business Architecture
  - Overview
  - Component Responsibilities
  - Application Models
- Client Development
  - UI Design Considerations
  - Components
- Connectivity
- Transcoding
- Demo: Inventory Tracking Application
  - Demo
  - Customized Architecture
- Summary

PalmSource Conference 2000

©IBM

Architecture Overview



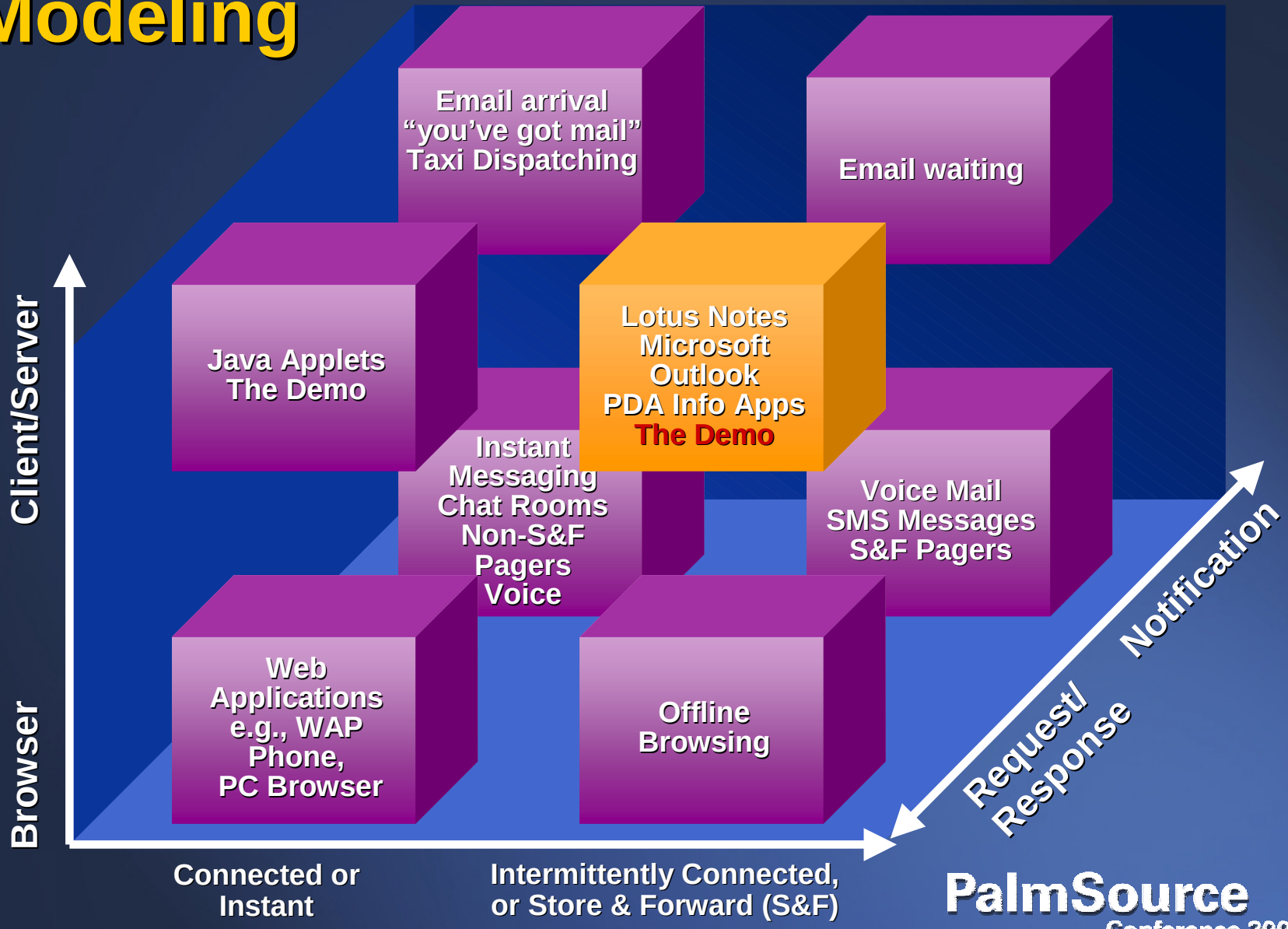
©IBM

Defining the Application Model

- We break down the problem of a pervasive architecture into 3 dimensions
  - How do you communicate to/from the client, i.e., Communication Model?
    - Request/Response (Pull) vs. Notification (Push)
  - What interface is required on the client, i.e., UI Model?
    - Client Server vs. Browser based
  - When does the client connect to the server, i.e., Connectivity Model?
    - Connected vs. Intermittently Connected

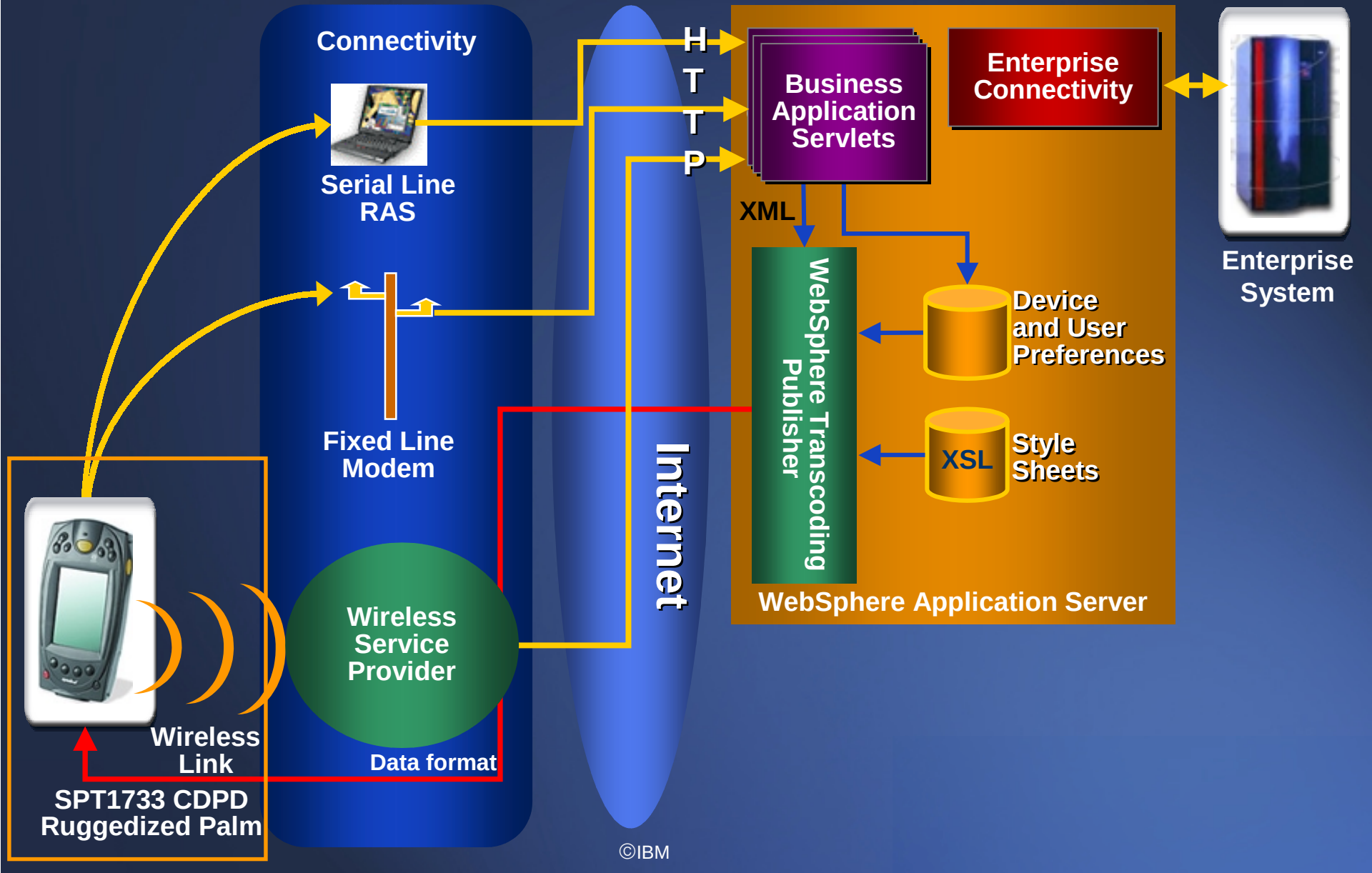
©IBM

The 3 Dimensions of Application Modeling



©IBM

Architecture—Client



©IBM

Symbol SPT1733



- **Palm OS®**
- **Built-in bar code scanner**
- **Rugged case—Can be dropped from 4 feet**
- **Built-in CDPD wireless modem**
- Pull capability
- 8 Meg storage capacity
- Has a land-line modem alternative if CDPD wireless unavailable
- Custom client can be written in C, C++, and Java
- Database capability
- Web-browser capable: WAP, HTML
- Software distribution Mgmt and any syncing handled by IBM Mobile Connect
- IR possible to mobile printer
- *Does not have push capability*

©IBM

Client Development

UI Design Considerations

- Minimize manual entry
- Fast Scan
- Custom keyboards and number pads—reduce error input
- Conversation box
- Functional tabs vs. directing the user
- Provide user hint on field format
- Popup lists vs. text fields
- Set focus to the field that is next in data entry

©IBM

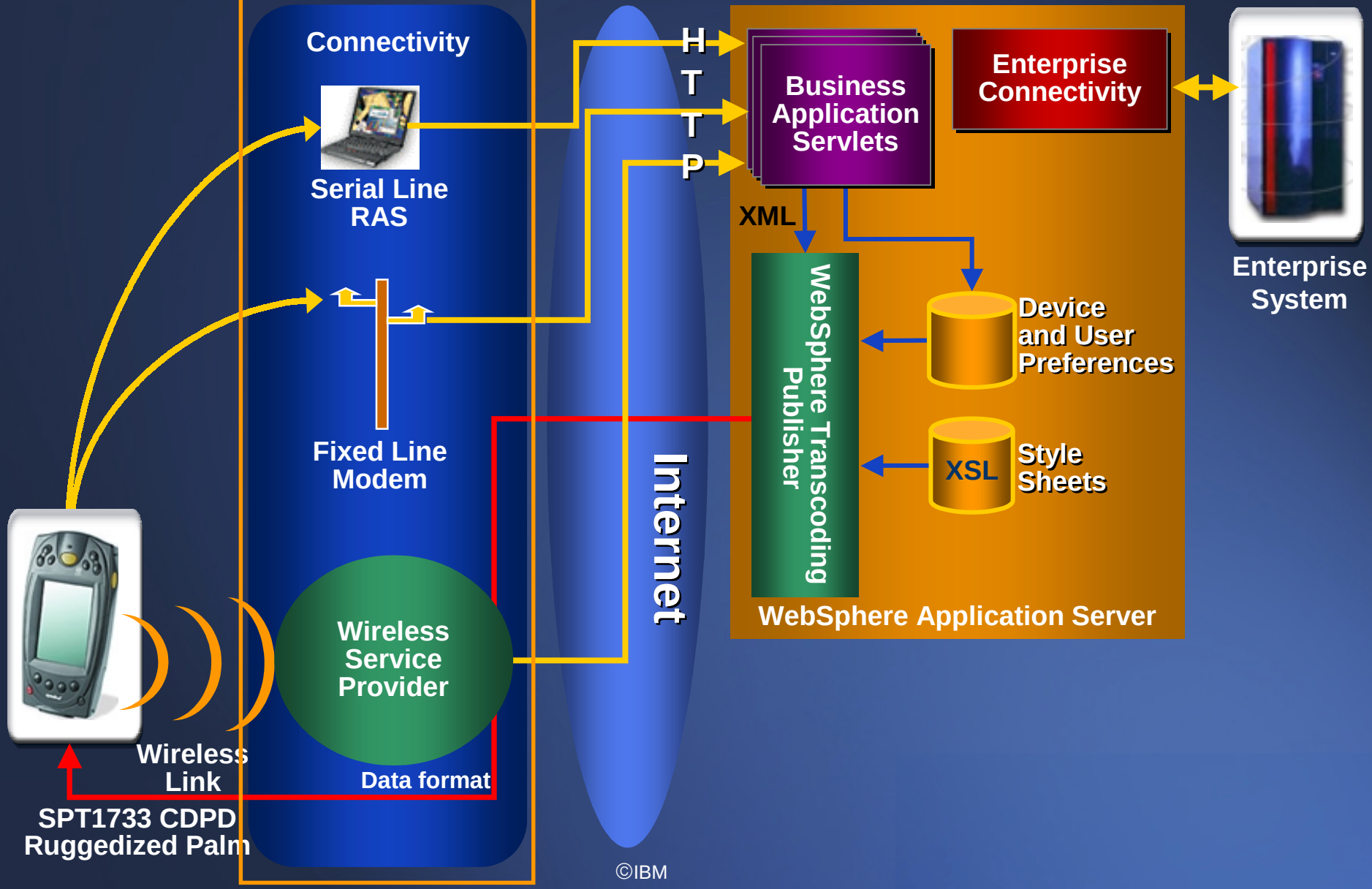
Client Development

Components

- Use cases
- Functional flow represented on charts
- IR Print
- Scanner libraries
- HTTP libraries

©IBM

Architecture—Connectivity



©IBM

Connectivity

- Wireless Gateway
  - No Infrastructure
  - Greater Mobility during data collection
  - But, Coverage Issues

©IBM

NATIONAL WIRELESS IP (CDPD) COVERAGE



PalmSource Conference 2000

©IBM

Connectivity

- Wireless Gateway
- Fixed Line: Modem
  - Needs an Analog line
- Serial Line: Remote Access Service, Conduit development
  - Needs a computer

©IBM

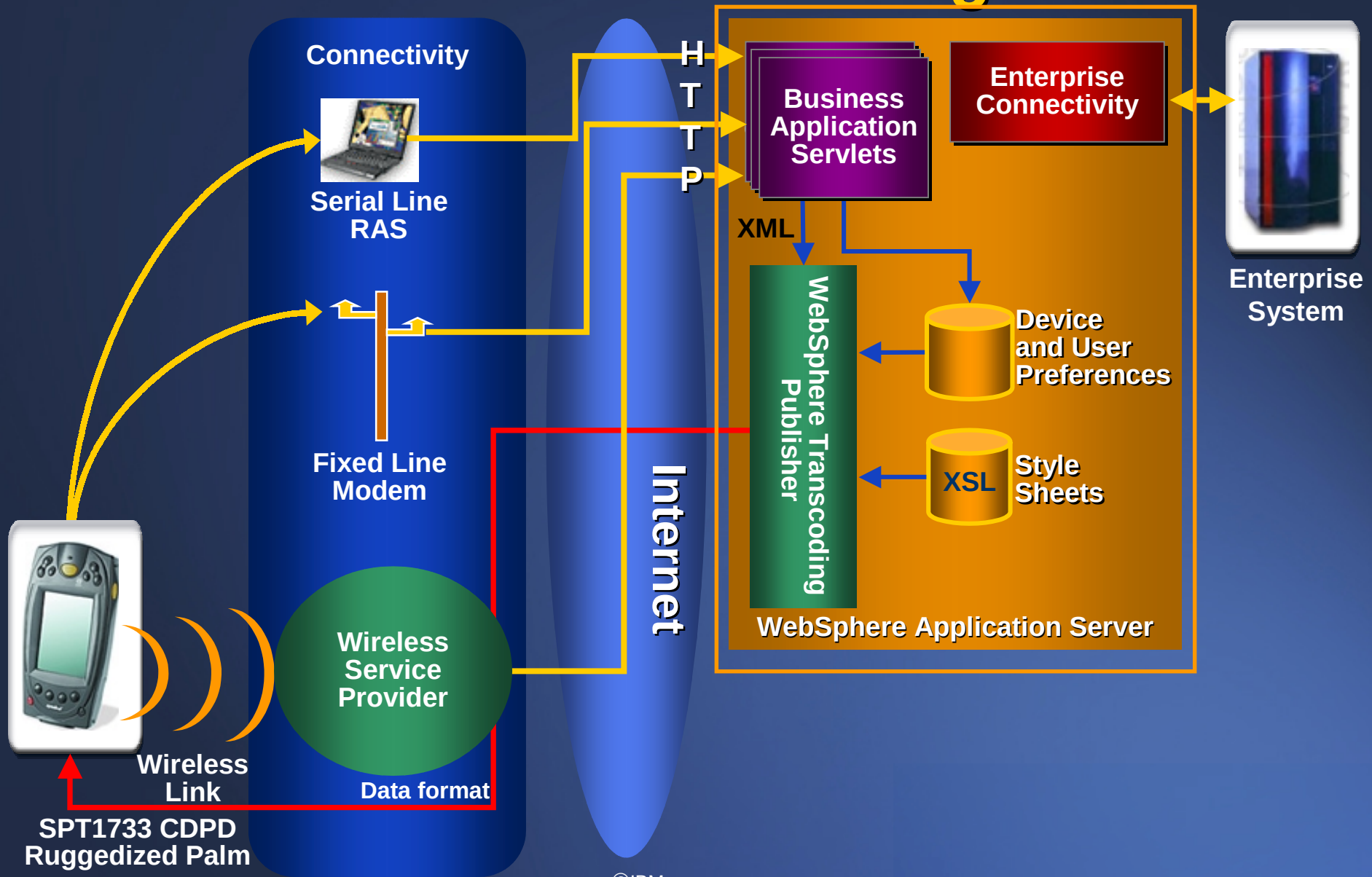
Connectivity—Protocol

What do I send and receive?

- Send: HTTP (Post, Get)
- Receive: NOT HTML, Blob (serialized string)
  - So how do we send blobs back?

©IBM

Architecture—Transcoding



©IBM

What Is “Transcoding”?

- ***Transcoding*** is the process of converting information from one format into another according to a well-defined algorithm or set of rules
- Examples:
  - Conversion of images from one encoding into another,
  - Conversion of HTML into speech markup language,
  - Conversion of one XML dialect into another
- The same output can be transcoded for output to standard desktop web browsers, smart phones, various types of PDAs, as well as devices using speech markup language

©IBM

How Does “Transcoding” Work?

INPUT XML

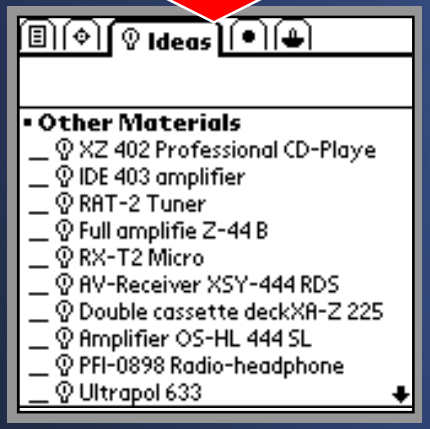
```
<ProductCatalog  Number = “WB0001” Variant =“2”>
<Item Material=“R-1001” Description = “XZ 402 Professional CD Player” Price=“123.99”/>
<Item Material=“R-1002” Description = “IDE 403 amplifier” Price=“172.32” />
<Item Material=“R-1003” Description = “RAT -2 Tuner ” Price=“199.99” />
</ProductCatalog>
```

PALM XSL Stylesheet

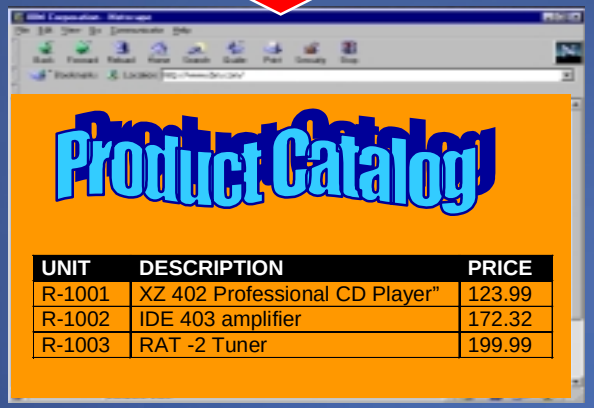
```
<xsl: value-of select= “@Description” />
<xsl:text>^</xsl:text>
<xsl: value-of select= “@Price” />
```

HTML Stylesheet

```
<TABLE > <TR>
<TD><xsl: value-of select= “@Description” /></TD>
<TD><xsl: value-of select= “@Price” /></TD>
</TR></TABLE>
```



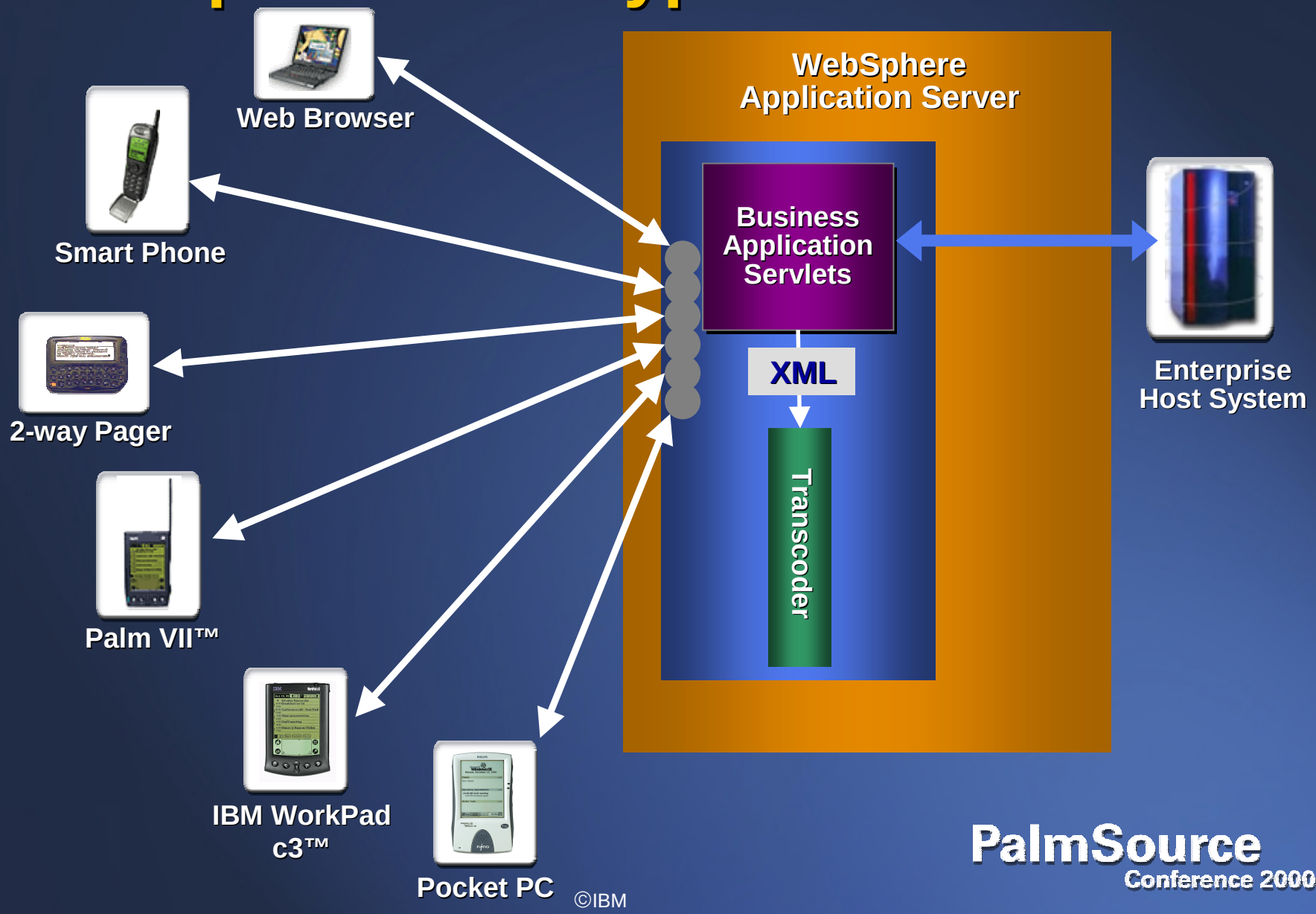
Palm



Web Browser

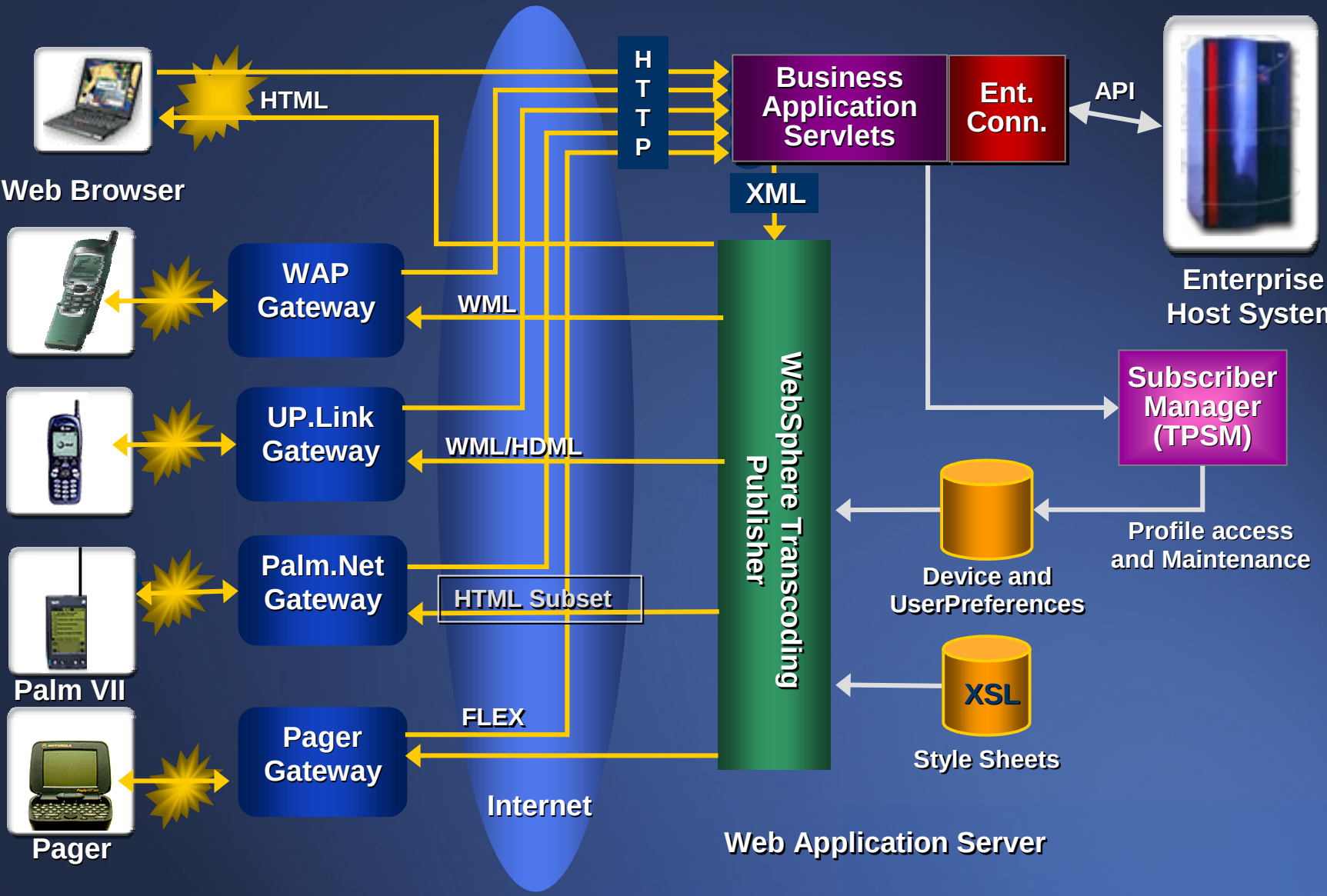
©IBM

Multiple Device Types



©IBM

Overall Architecture



©IBM



Demo

PalmSource

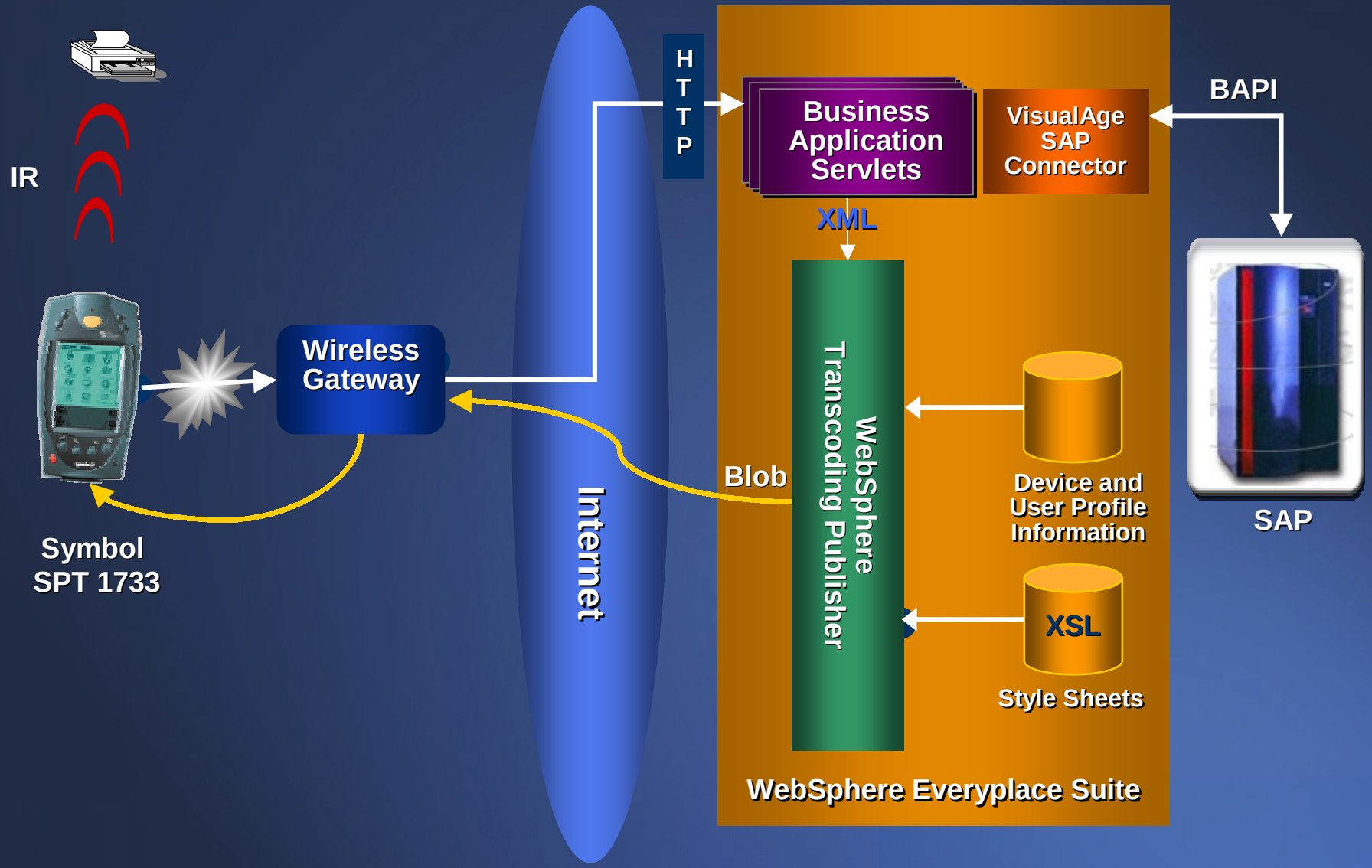
Conference 2000

Demo: Inventory Tracking Application

- Definitions
  - Boxes and Pallets
  - Goods Received (GR)
  - Goods Issue (GI)
- Data collection
- IR Printing
- Data transmission

©IBM

Demo Solution Architecture



©IBM

Architecture Component Responsibilities

Client Device	Gateway	Application Servlet	Transcoding Publisher	Profile Management	Enterprise Server
Present UI appropriate to channel	Channel-specific connectivity	Application flow and front-end logic	Reformatting of response pages	Channel/user profile maintenance	Business logic and client data
Uses browser for connected applications	Turns user request into HTTP(S)	HTTP(S) in, XML out	XML in, HTML/WML/ HDML/XML out	Used for user authentication and authorization	Accessed through Enterprise Connector
Custom UI for client-server applications	Physical transmission of response to device	Generic code, independent of channel	Customize output for each channel/user	Defines which channel(s) to use for push messages	Provide common set of business transactions

©IBM

Summary

- IBM's mobile e-business Architecture
  - Application Models
  - Offerings in Travel, Banking, Distribution, Retail, and other industries
- Client Development
  - UI Design considerations
- Connectivity: 3 models
- Transcoding
- Demo: Sample Tracking Application
- mutukuru@us.ibm.com

PalmSource Conference 2000

©IBM



Developing Java™ Apps on Palm OS Using IBM VisualAge Micro Edition

Chris Laffra
Object Technology International

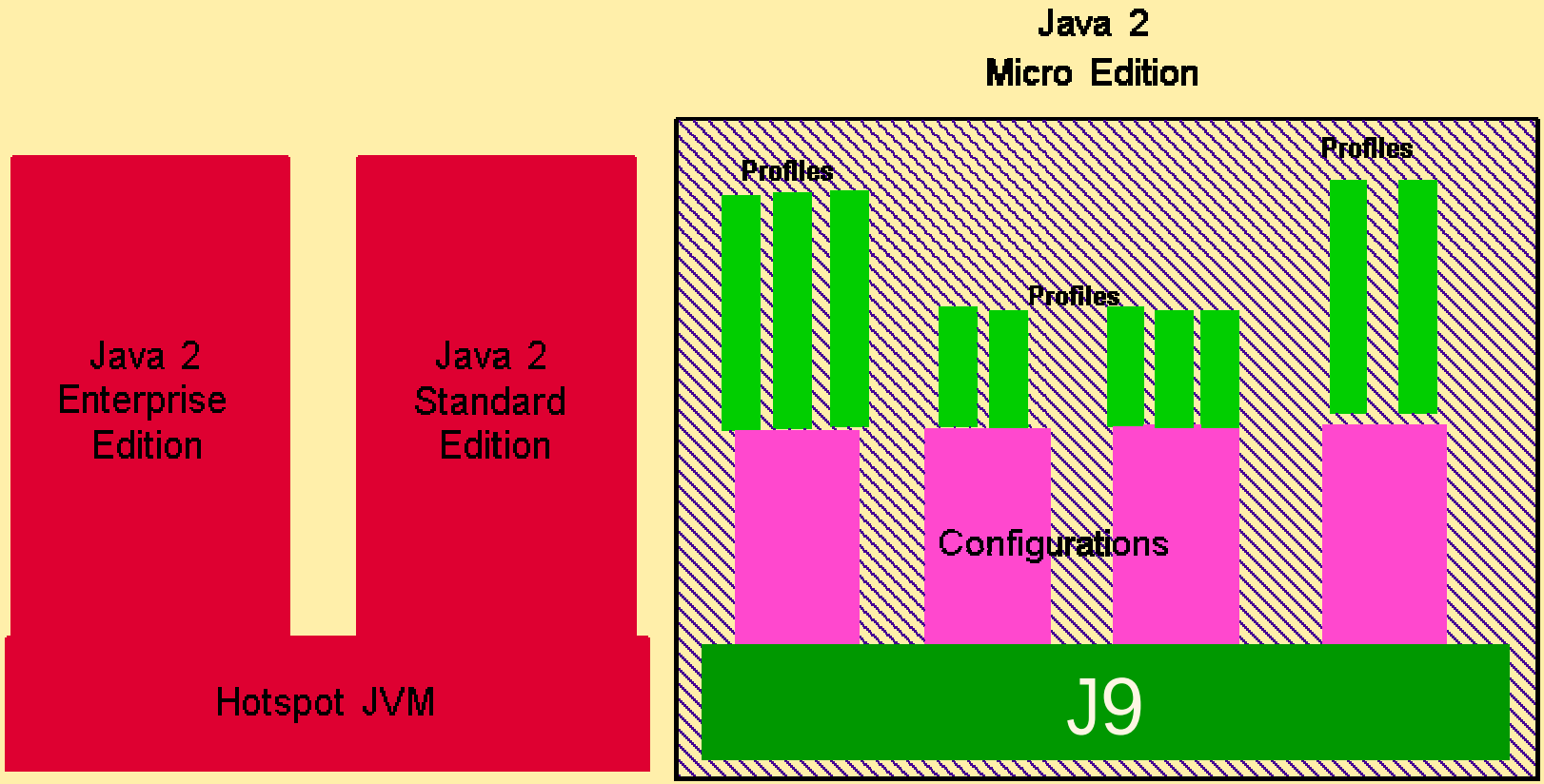
PalmSource Conference 2000

OTI Positioning

- OTI wholly owned of IBM since few years
- 200 people worldwide (Canada/US/Europe)
  - VA Smalltalk, VAJava, VA Micro Edition
- OTI's common code base strategy:
  - Modular
  - Portable
  - Scalable
  - Elegant
  - Independent

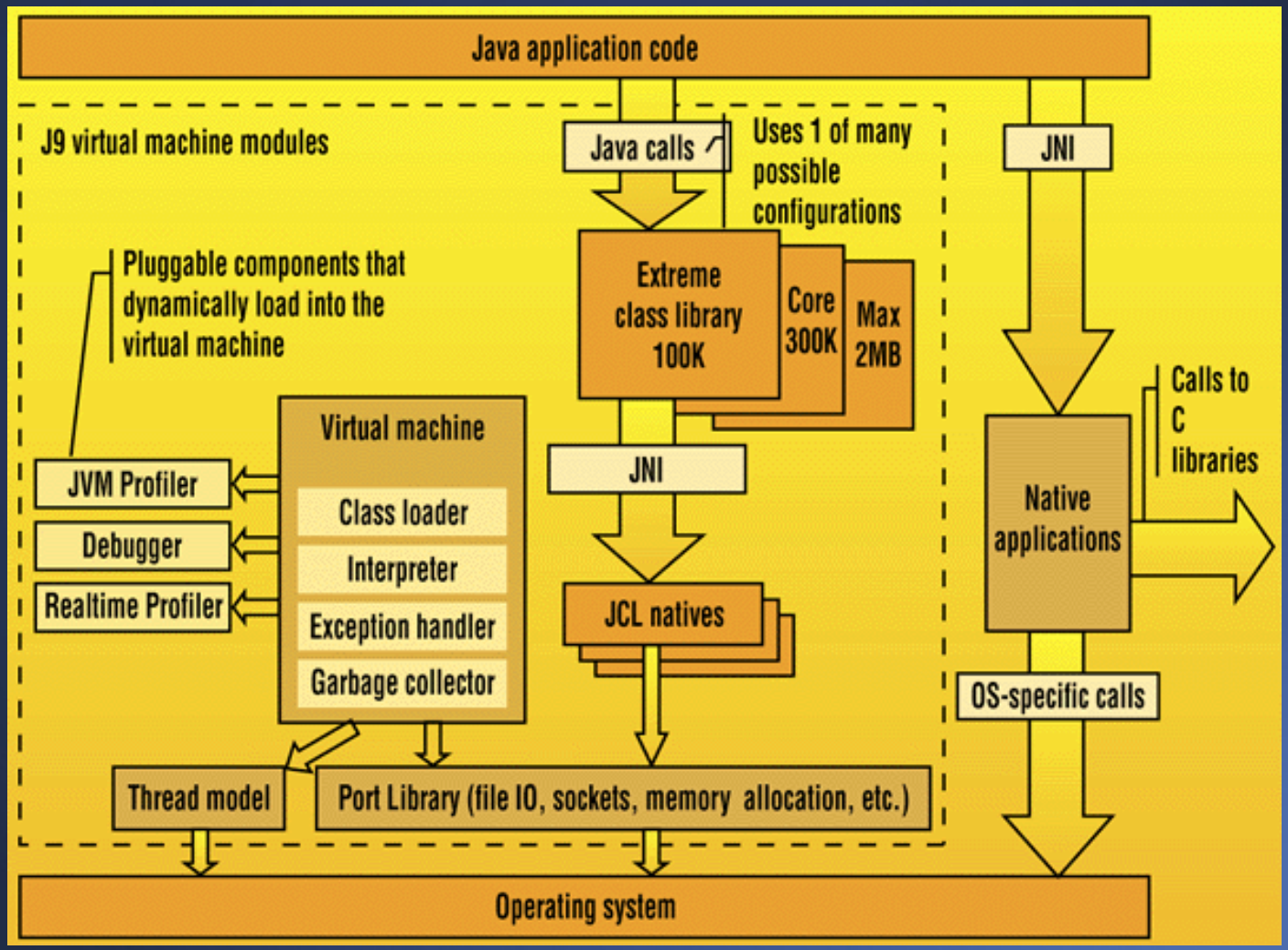
©IBM

VM Landscape (Simple Story)



©IBM

J9, Common Code Base Strategy



PalmSource Conference 2000

©IBM

The Palm OS® Platform

- Palm OS (3.0, 3.2, 3.3, 3.5, …), Motorola 68K
- Palm Handhelds (Palm III, Palm V, Palm VII)
- Lots of “challenges”:
  - Small screen (typical: 160 x 160)
  - Slow screen (LCD), B&W or low color
  - Relatively small amounts of memory
  - Lots of technical details (high geek factor)
  - Main programming interface: C
  - No portability



PalmSource Conference 2000

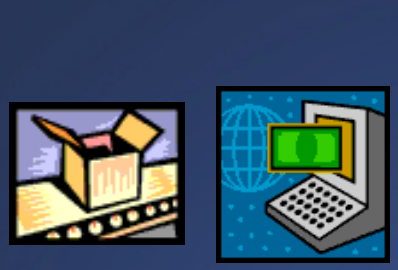
©IBM

Java Solution

- Free VisualAge Micro Edition Personal Edition IDE (see CD distributed at conference)
- Free Palm OS runtime download, no runtime fees (http://www.ibm.com/embedded)
- Porting effort started February 2000
- Distributed development team:
  - Ottawa: VM + base class libraries
  - Amsterdam: Palm libs + samples + release build
  - Phoenix: Release building

©IBM

Java™ 2 Platform



???



Profiles

Configu-rations

Pro File | Pro File

Pro File | Pro File

Pro File | Pro File

J2EE APIs

J2SE APIs

J2ME APIs

Java Virtual Machine

Hardware / Operating System

PalmSource Conference 2000

©IBM

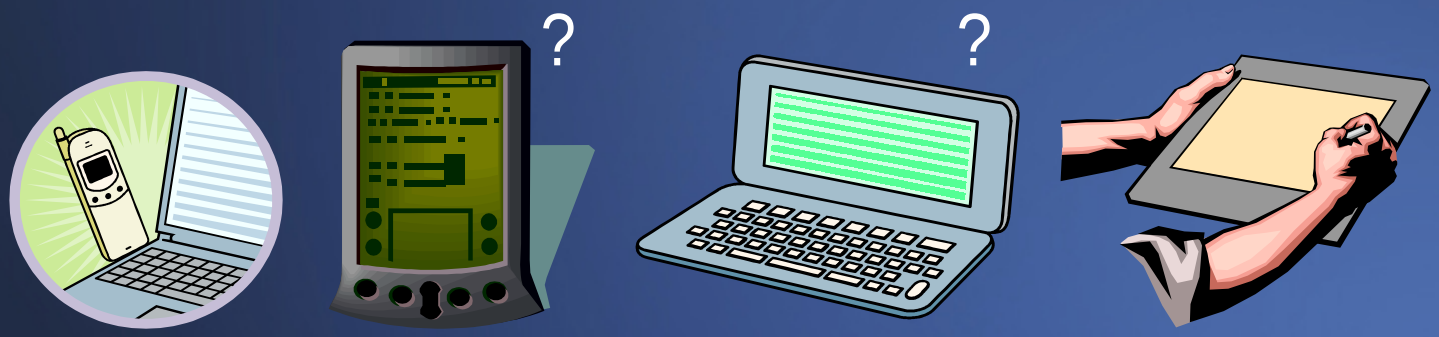
Configuration: CLDC

- Connected Limited Device Configuration
- JSR-000030
- **Quick Overview:**
  - Small Memory: 256K–512K
  - Limited power (battery)
  - Some type of connectivity (maybe < 9600Bps)
  - No or limited UI

©IBM

CLDC Profile: MID

- MID (Mobile Information Device) Profile
- Based on CLDC
- JSR-000037
- Very limited in scope: think cell-phones



©IBM

CLDC Profile: PDAP

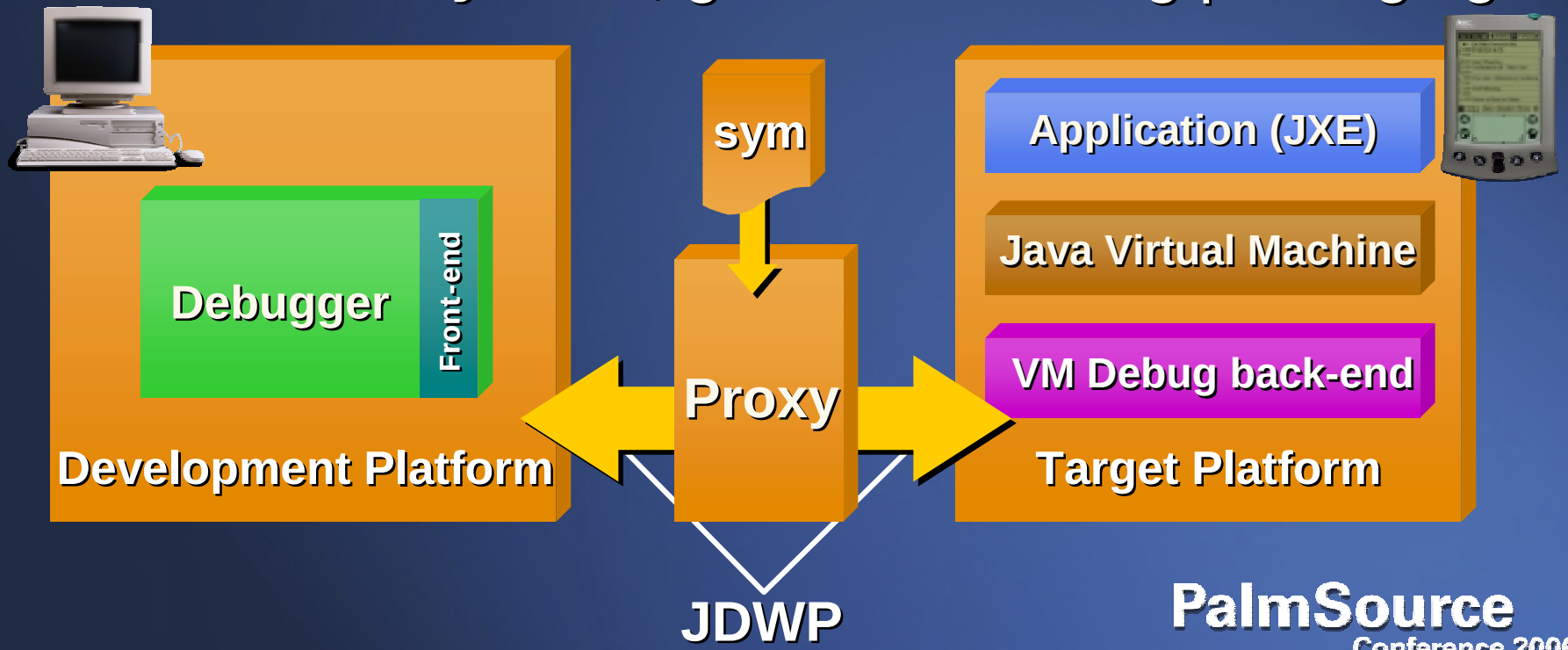
- PDA (Personal Digital Assistent) Profile
- Based on CLDC/MIDP
- JSR-000075
- Less limited in scope: think subset of AWT



©IBM

Remote Symbolic Debugging

- Symbolic debug info takes lots of space
- Cannot be stored at embedded device
- Saved in **sym** file; generated during packaging



©IBM

Demonstration Highlights

- Palm OS abstraction classes:
  - One Java class for each C struct
  - One method in class OS for each C function
- Resources, dynamic UI, kJava and AWT
- build+link+import+run, in one single command
- End of VM cleanup
- C memory (chunk) leakage prevention
- Debugging

©IBM


accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000